Mark schemes

Q1.

Marking Instructions	AO	Marks	Typical Solution
Correctly states that the student's argument is invalid because the integrand is undefined at a value in the interval $(0, \frac{\pi}{2})$	AO2.3	E1	When $x = \frac{\pi}{4}$, cos x – sin x = 0
Correctly justifies the reason why it is undefined and states a correct conclusion	AO2.4	E1	∴ the integrand is undefined at this point and the integral *is* improper
			Total 2 marks

Q2.

Marking Instructions	AO	Marks	Typical Solution
Splits integrand into partial fractions of the form $\frac{Ax+B}{x^2+5}+\frac{C}{3x+2}$	AO3.1a	M1	$\frac{4x-30}{(x^2+5)(3x+2)} \equiv \frac{Ax+B}{x^2+5}+\frac{C}{3x+2}$
Sets up an identity from which to solve for A, B and C	AO1.1a	M1	$\Rightarrow 4x - 30 \equiv (Ax + B)(3x + 2) + C(x^2 + 5)$
Obtains correct values of A, B and C CAO	AO1.1b	A1	Compare coefficients of x: 4 = 2A + 3B Compare coefficients of x^2: 0 = 3A + C
Integrates 'their' two terms correctly FT provided both M1 marks awarded	AO1.1b	A1F	Compare constant terms: –30 = 2B + 5C ∴ –30 = 2B – 15A $-30 = 2B - 15(\frac{4-3B}{2})$
Applies the laws of logs to 'their' integral correctly	AO1.1a	M1	B = 0 A = 2 C = –6
Applies limits (a and 0) to 'their' integral correctly	AO1.1a	M1	
Shows the limiting process used with clear detailed working	AO2.1	R1	

Obtains correct single term solution CAO	AO1.1b	A1	$\int \frac{4x-30}{(x^2+5)(3x+2)}\mathrm{d}x$ $= \int \frac{2x}{x^2+5} - \frac{6}{3x+2}\mathrm{d}x$ $= \ln(x^2 + 5) - 2\ln(3x + 2) + c$ $\int_0^\infty \frac{4x-30}{(x^2+5)(3x+2)}\mathrm{d}x$ $= \lim_{\alpha\to\infty} \int_0^a \frac{2x}{x^2+5} - \frac{6}{3x+2}\mathrm{d}x$ $= \lim_{\alpha\to\infty} \left[\ln(x^2+5) - 2\ln(3x+2)\right]_0^a$ $= \lim_{a\to\infty} \left[\ln\left(\frac{x^2+5}{(3x+2)^2}\right)\right]_0^a$ $= \lim_{a\to\infty} \left[\ln\left(\frac{a^2+5}{9a^2+12x+4}\right) - \ln\left(\frac{5}{4}\right)\right]$ $= \ln\left(\frac{1}{9}\right) - \ln\left(\frac{5}{4}\right)$ $= \ln\left(\frac{4}{45}\right)$
			Total 8 marks

Q3.

$\int \frac{1}{x\sqrt{x}}\,\mathrm{d}x = \int x^{-\frac{3}{2}}\,(\mathrm{d}x)$

$\int x^{-\frac{3}{2}}$ PI

M1

$= -2x^{-\frac{1}{2}}$ $(+c)$

ACF, can be unsimplified.
Condone absence of $+c$

A1

$-2x^{-\frac{1}{2}} \to 0$ as $x \to \infty$

OE Ft on kx^{-n}, $n > 0$

E1

$\int_{25}^{\infty} \frac{1}{x\sqrt{x}}\,\mathrm{d}x = \frac{2}{5}$

A1

[4]

Q4.

(a) Integrand is not defined at $x = 0$

OE

E1

1

(b) $\int x^4 \ln x \, dx = \frac{x^5}{5} \ln x - \int \frac{x^5}{5}\left(\frac{1}{x}\right) dx$

......$= kx^5 \ln x \pm \int f(x)$, with $f(x)$ not involving the 'original' $\ln x$

M1

A1

...... $= \frac{x^5}{5} \ln x - \frac{x^5}{25}$ $(+ c)$

A1

$\int_0^1 x^4 \ln x \, dx = \{ \lim_{a \to 0} \int_a^1 x^4 \ln x \, dx \} = -\frac{1}{25} - \lim_{a \to 0} [\frac{a^5}{5}\ln a - \frac{a^5}{25}]$

Limit 0 replaced by a limiting process and F(1) − F(a) OE

M1

But $\lim_{a \to 0} a^5 \ln a = 0$

Accept $\lim_{x \to 0} x^k \ln x = 0$ for any $k > 0$

E1

So $\int_0^1 x^4 \ln x \, dx = -\frac{1}{25}$

Dep on M and A marks all scored

A1

6

[7]

Q5.

$I = \int_1^\infty x^{-4}(2x^2 - 5x) \, dx = \int_1^\infty (2x^{-2} - 5x^{-3}) \, dx$

$I = \left[-2x^{-1} - 5\frac{x^{-2}}{-2} \right]_1^\infty$

At least one term correct

M1

As $x \to \infty$, $x^{-1} \to 0$ and $x^{-2} \to 0$

OE Ft on kx^{-n} provided M1 awarded

E1

$I = 0 - (-2 + 5/2) = -\frac{1}{2}$

(I =) $-\frac{1}{2}$ Dep on both terms integrated correctly in the M1 line

A1

[3]

Q6.

$\int \left(\frac{2x}{x^2+4} - \frac{4}{2x+3}\right) dx = \ln(x^2 + 4)$

B1

$- 2\ln(2x + 3)$ $\{+c\}$

OE

B1

(I =) $\lim_{a\to\infty} \int_0^a \left(\frac{2x}{x^2+4} - \frac{4}{2x+3}\right) dx$

∞ replaced by a (OE) and $\lim_{a\to\infty}$ seen or taken at any stage

M1

$= \lim_{a\to\infty} \left[\ln(x^2+4) - 2\ln(2x+3)\right]_0^a$

***Remaining marks are dep** on getting*
$p\ln(x^2 + 4) + q\ln(2x + 3)$
after integration, where p and q are non-zero constants

$= \lim_{a\to\infty} \left[\ln(a^2+4) - 2\ln(2a+3)\right] - (\ln 4 - 2\ln 3)$

Dealing with the 0 limit correctly and using
$\ln P - \ln Q = \ln(P/Q)$ at least once at any stage
either before or after using F( )−F(0).
OE

M1

$= \lim_{a\to\infty} \left[\ln\left(\frac{a^2+4}{(2a+3)^2}\right)\right] - (\ln 4 - \ln 9)$

OE

$$= \lim_{a \to \infty} \left[\ln\left(\frac{1 + \frac{4}{a^2}}{4 + \frac{12}{a} + \frac{9}{a^2}} \right) \right] - (\ln 4 - \ln 9)$$

Writing F(a) OE in a suitable form when considering $a \to \infty$ OE

M1

$$\mathrm{I} = \int_0^\infty \left(\frac{2x}{x^2+4} - \frac{4}{2x+3} \right) \mathrm{d}x = \ln\frac{1}{4} - \ln\frac{4}{9} = \ln\frac{9}{16}$$

CSO

A1

[6]

Q7.

(a) $\int x^{-2/3}\,\mathrm{d}x = 3x^{1/3}$ $(+c)$

$kx^{\frac{1}{3}}$, $k \neq 0$ ie condone incorrect non-zero coefficient here

B1

$(3)x^{1/3}$ → Ý as x → Ý, so no finite value

E1

(b) $\int x^{-4/3}\,\mathrm{d}x = -3x^{-1/3}$ $(+c)$

$\lambda x^{-1/3}$, $\lambda \neq 0$

M1

$-3x^{-1/3}$ OE

A1

$\int_8^\infty x^{-4/3}\,\mathrm{d}x = -3(0 - \tfrac{1}{2}) = \tfrac{3}{2}$

CSO

A1

[5]

Q8.

(a) The interval of integration is infinite

OE

E1

1

(b) $u = x^2 e^{-4x} + 3 \Rightarrow \mathrm{d}u = (2xe^{-4x} - 4x^2e^{-4x})\,\mathrm{d}x$

du / dx or 'better'

M1

$\int \frac{x(1-2x)}{x^2+3e^{4x}} dx = \int \frac{1}{2} \times \frac{2x(1-2x)e^{-4x}}{x^2e^{-4x}+3} dx = \frac{1}{2} \times \int \frac{1}{u} du$

A1

$= \frac{1}{2} \ln u + c = \frac{1}{2} \ln(x^2e^{-4x} + 3) \quad \{+c\}$

OE Condone missing c.
Accept later substitution back if explicit

A1

3

(c) $I = \int_{\frac{1}{2}}^{\infty} \frac{x(1-2x)}{x^2+3e^{4x}} dx = \lim_{a\to\infty} \int_{\frac{1}{2}}^{a} \frac{x(1-2x)}{x^2+3e^{4x}} dx$

M1

$\lim_{a\to\infty} \frac{1}{2}\{\ln (a^2e^{-4a} + 3) - \ln(\frac{e^{-2}}{4} + 3)\}$

Uses part (b) and F(a) − F(1/2)

M1

$= \frac{1}{2}\ln\{\lim_{a\to\infty}(a^2e^{-4a} + 3)\} - \frac{1}{2}\ln(\frac{e^{-2}}{4} + 3)$

Now $\lim_{a\to\infty}(a^2e^{-4x}) = 0$

Stated explicitly (could be in general form)

E1

$I = \frac{1}{2}\ln 3 - \frac{1}{2}\ln(\frac{e^{-2}}{4} + 3)$

CSO ACF

A1

4

[8]

Q9.

(a) $\int x^2e^{-x} dx = -x^2e^{-x} - \int -2xe^{-x} dx$

$kx^2e^{-x} - \int kxe^{-x}$ (dx) for $k = \pm 1$

M1
A1

$= -x^2e^{-x} + 2\{-xe^{-x} - \int -e^{-x}dx\}$

$\int xe^{-x}\,dx = \lambda xe^{-x} - \int \lambda e^{-x}(dx)$ for $\lambda = \pm 1$ in 2nd application of integration by parts

m1

$= -x^2e^{-x} - 2xe^{-x} - 2e^{-x}\ (+c)$

Condone absence of + c

A1

4

(b) $\mathrm{I} = \int_0^\infty x^2e^{--}\,dx = \lim_{a \to \infty} \int_0^a x^2e^{-x}\,dx$

F(a) − F(0) with an indication of limit '$a \to \infty$' and F(x) containing at least one x^ne^{-x}, $n > 0$ term

M1

$\lim_{a \to \infty} \{-a^2e^{-a} - 2ae^{-a} - 2e^{-a}\} - [-2]$

$\lim_{a \to \infty} a^k e^{-a} = 0, \qquad (k > 0)$

For general statement or specific statement for either $k = 1$ or $k = 2$

E1

$\int_0^\infty x^2e^{-x}\,dx = 2$

CSO

A1

2

[6]

Q10.

(a) $\int 2x^{-3}\,dx = -x^{-2}\ (+c)$

M1 for correct index

M1A1

$\int_p^q 2x^{-3}\,dx = p^{-2} - q^{-2}$

OE; ft wrong coefficient of x^{-2}

A1F

3

(b) (i) As $p \to 0$, $p^{-2} \to \infty$, so no value

B1

(ii) As $q \to \infty$, $q^{-2} \to 0$, so value is ¼

ft wrong coefficient of x^{-2}
or reversal of limits

M1A1F

3

[6]

Q11.

(a) $$\frac{12x+8-12x-3}{(4x+1)(3x+2)} = \frac{5}{(4x+1)(3x+2)}$$

Accept $C = 5$

B1

1

(b) $$\int \frac{10}{(4x+1)(3x+2)}\,dx = 2\int\left(\frac{4}{(4x+1)} - \frac{3}{(3x+2)}\right)dx$$

M1

$$= 2[\ln(4x+1) - \ln(3x+2)]\ (+c)$$

OE

A1

$$I = \lim_{a\to\infty}\int_1^a\left(\frac{10}{(4x+1)(3x+2)}\right)dx$$

∞ replaced by a and $\lim_{a\to\infty}$ *(OE)*

M1

$$= 2\lim_{a\to\infty}[\ln(4a+1) - \ln(3a+2)] - (\ln 5 - \ln 5)$$

$$= 2\lim_{a\to\infty}\left[\ln\left(\frac{4a+1}{3a+2}\right)\right] = 2\lim_{a\to\infty}\left[\ln\left(\frac{4+\frac{1}{a}}{3+\frac{2}{a}}\right)\right]$$

Limiting process shown.
Dependent on the previous M1M1

m1,m1

$= 2\ln\frac{4}{3} = \ln\frac{16}{9}$

CSO

A1

6

[7]

Q12.

(a) $\int x^2 \ln x\,dx = \frac{x^3}{3}\ln x - \int \frac{x^3}{3}\left(\frac{1}{x}\right)dx$

... $= kx^3 \ln x \pm \int f(x)$, *with f(x) not involving the 'original' ln x*

M1

A1

...... $= \frac{x^3}{3}\ln x - \frac{x^3}{9}(+c)$

Condone absence of '+ c'

A1

3

(b) Integrand is not defined at $x = 0$

OE

E1

1

(c) $\int_0^e x^2 \ln x\,dx = \left\{\lim_{a\to 0}\int_a^e x^2 \ln x\,dx\right\}$

OE

$= \left(\frac{e^3}{3}\ln e - \frac{e^3}{9}\right) - \lim_{a\to 0}\left[\frac{a^3}{3}\ln a - \frac{a^3}{9}\right]$

F(e) − $\lim_{a\to 0}$ *[F(a)]*

M1

But $\lim_{a\to 0} a^3 \ln a = 0$

Accept a general form eg $\lim_{a\to 0} x^k \ln x = 0$

E1

So $\int_0^e x^2 \ln x \, dx = \frac{2e^3}{9}$

CSO

A1

3

[7]

Q13.

(a) $x^{-\frac{1}{2}} \to \infty$ as $x \to 0$

Condone "$x^{-\frac{1}{2}}$ has no value at x = 0"

(b) (i) $\int x^{-\frac{1}{2}} dx = 2x^{\frac{1}{2}} \; (+c)$

M1 for correct power of x

M1A1

$$\int_0^{\frac{1}{16}} x^{-\frac{1}{2}} dx = \frac{1}{2}$$

ft wrong coefficient of $x^{\frac{1}{2}}$

A1F

3

(ii) $\int x^{-\frac{5}{4}} dx = -4x^{-\frac{1}{4}} \; (+c)$

M1 for correct power of x

M1A1

$x^{-\frac{1}{4}} \to \infty$ as $x \to 0$, so no value

ft wrong coefficient of $x^{-\frac{1}{4}}$

E1F

3

[7]

Q14.

(a) The interval of integration is infinite

OE

E1

1

(b) (i) $x = \frac{1}{y} \Rightarrow' dx = -y^{-2} \; dy'$

$$\int \frac{\ln x^2}{x^3}\,dx \Rightarrow \int (y^3 \ln y^{-2})(-y^{-2})\,dy$$

M1

$$= \int -y \ln y\,dy = \int 2y \ln y\,dy$$

CSO *AG*

A1

2

(ii) $$\int 2y \ln y\,dy = y^2 \ln y - \int y^2\left(\frac{1}{y}\right)dy$$

...= $ky^2 \ln y \pm \int f(y)\,dy$ *with f(y) not involving the 'original' ln y*

M1A1

$$\ldots\ldots = y^2 \ln y - \frac{1}{2}y^2 + c$$

Condone absence of '+ c'

A1

$$\int_0^1 2y \ln y\,dy = \lim_{a\to 0}\int_a^1 2y \ln y\,dy$$

$$= \left(0 - \frac{1}{2}\right) - \lim_{a\to 0}\left[a^2 \ln a - \frac{a^2}{2}\right]$$

M1

$$= -\frac{1}{2} \text{ since } \lim_{a\to 0} a^2 \ln a = 0$$

CSO Must see clear indication that cand has correctly considered

$$\lim_{a\to 0} a^k \ln a = 0$$

A1

5

(iii) $$\text{So } \int_1^{\infty} \frac{\ln x^2}{x^3}\,dx = \frac{1}{2}$$

ft on minus c's value as answer to (b)(ii)

B1F

1

[9]

Q15.

(a) The interval of integration is infinite

OE

E1

1

(b) $\int 4xe^{-4x}\,dx = -xe^{-4x} - \int -e^{-4x}\,dx$

$kxe^{-4x} - \int ke^{-4x}\,dx$ *for non-zero k*

M1A1

$= -xe^{-4x} - \frac{1}{4}e^{-4x}\{+c\}$

Condone absence of + c

A1F

3

(c) $I = \int_1^{\infty} 4xe^{-4x}\,dx = \lim_{a\to\infty}\int_1^{a} 4xe^{-4x}\,dx$

$\lim_{a\to\infty}\{-ae^{-4a} - \frac{1}{4}e^{-4a}\} - \left[-\frac{5}{4}e^{-4}\right]$

F(a) – F(1) with an indication of limit 'a → ∞'

M1

$\lim_{a\to\infty} a\,e^{-4a} = 0$

For statement with limit/ limiting process shown

M1

$I = \frac{5}{4}e^{-4}$

CSO

A1

3

[7]

Q16.

(a) $\int x^{\frac{3}{4}}\,dx = 4x^{\frac{1}{4}}(+c)$

M1 if index correct

M1A1

This tends to ∞ as $x \to \infty$, so no value

ft wrong coefficient

A1F

3

(b) $\int x^{-\frac{5}{4}}\, dx = -4x^{-\frac{1}{4}} (+c)$

M1 if index correct

M1A1

$\int_1^{\infty} x^{-\frac{5}{4}}\, dx = 0 - (-4) = 4$

ft wrong coefficient

A1F

3

(c) Subtracting 4 leaves ∞, so no value

ft if c has 'no value' in (a) but has a finite answer in (b)

B1F

1

[7]

Q17.

(a) $\int \ln x\, dx = x \ln x - \int x\left(\frac{1}{x}\right) dx$

Integration by parts

M1

$= x \ln x - x + c$

CSO *AG*

A1

2

(b) $\int_0^1 \ln x\, dx = \lim_{a\to 0} \int_a^1 \ln x\, dx$

OE

M1

$= \lim_{a\to 0} \{0 - 1 - [a \ln a - a]\}$

F(1) – F(a) OE

M1

But $\lim_{a \to 0} a \ln a = 0$

Accept a general form eg

$\lim_{a \to 0} a^k \ln a = 0$

E1

So $\int_0^1 \ln x \, dx = -1$

A1

4

[6]

Q18.

$$\int \left(\frac{1}{x} - \frac{4}{4x+1} \right) dx = \ln x - \ln (4x+1) \{+c\}$$

OE

B1

$$I = \lim_{a \to \infty} \int_1^a \left(\frac{1}{x} - \frac{4}{4x+1} \right) dx$$

∞ *replaced by a (OE) and* $\lim_{a \to \infty}$

M1

$$= \lim_{a \to \infty} [\ln x - \ln (4x+1)]_1^a$$

$$= \lim_{a \to \infty} \left[\ln \left(\frac{1}{4a+1} \right) - \ln \frac{1}{5} \right]$$

$\ln a - \ln (4a+1) = \ln \left(\frac{a}{4a+1} \right)$

***and** previous M1 scored*

m1

$$= \lim_{a \to \infty} \left[\ln \left(\frac{1}{4 + \frac{1}{a}} \right) - \ln \frac{1}{5} \right]$$

$\ln \left(\frac{a}{4a+1} \right) = \ln \left(\frac{1}{4 + \frac{1}{a}} \right)$ ***and***

previous M1m1 scored

m1

$$= \ln\frac{1}{4} - \ln\frac{1}{5} = \ln\frac{5}{4}$$

CSO

A1

[5]

Q19.

(a) The interval of integration is infinite

OE

E1

1

(b) $$\int xe^{-3x}\,dx = -\frac{1}{3}xe^{-3x} - \int -\frac{1}{3}e^{-3x}\,dx$$

Reasonable attempt at parts

M1A1

$$= -\frac{1}{3}xe^{-3x} - \frac{1}{9}e^{-3x}\{+c\}$$

Condone absence of + c

A1ft

3

(c) $$I = \int_1^{\infty} xe^{-3x}\,dx = \lim_{a\to\infty}\int_1^{a} xe^{-3x}\,dx$$

$$\lim_{a\to\infty}\{-\frac{1}{3}ae^{-3a} - \frac{1}{9}e^{-3a}\} - \left[-\frac{4}{9}e^{-3}\right]$$

F(a) – F(1) with an indication of limit 'a → ∞'

M1

$$\lim_{a\to\infty} ae^{-3a} = 0$$

For statement with limit/limiting process shown

M1

$$I = \frac{4}{9}e^{-3}$$

A1

3

[7]

Q20.

(a) $\int x^{-\frac{1}{2}} dx = 2x^{\frac{1}{2}} (+c)$

M1 for correct power in integral

M1A1

$x^{½} \to \infty$ as $x \to \infty$, so no value

E1

3

(b) $\int x^{-\frac{3}{2}} dx = -2x^{-\frac{1}{2}} (+c)$

M1 for correct power in integral

M1A1

$x^{-½} \to 0$ as $x \to \infty$

PI

E1

$$\int_9^\infty x^{-\frac{3}{2}} dx = -2(0 - \frac{1}{3}) = \frac{2}{3}$$

Allow A1 for correct answer even if not fully explained

A1

4

[7]

Q21.

(a) $\int x^3 \ln x \, dx = \frac{x^4}{4} \ln x - \int \frac{x^4}{4} \left(\frac{1}{x}\right) dx$

... = kx^4 ln x ±∫f(x), with f(x) not involving the 'original' ln x

M1A1

$$\ldots\ldots = \frac{x^4}{4} \ln x - \frac{x^4}{16} + c$$

Condone absence of '+ c'

A1

3

(b) Integrand is not defined at $x = 0$

OE

E1

1

(c) $\int_0^{e} x^3 \ln x \, dx = \left\{ \lim_{a \to 0} \int_a^{e} x^3 \ln x \, dx \right\}$

$= \frac{3e^4}{16} - \lim_{a \to 0} \left[\frac{a^4}{4} \ln a - \frac{a^4}{16} \right]$

F(e) – F(a)

M1

But $\lim_{a \to 0} a^4 \ln a = 0$

Accept a general form eg $\lim_{x \to 0} x^4 \ln x = 0$

B1

So $\int_0^{e} x^3$ ln *x* d*x* exists and = $\frac{3e^4}{16}$

CSO

A1

3

[7]

Q22.

(a) Integrand is not defined at *x* = 0

OE

E1

1

(b) $\int x^{\frac{1}{2}} \ln x \, dx = 2x^{\frac{1}{2}} \ln x - \int 2x^{\frac{1}{2}} \left(\frac{1}{x} \right) dx$

M1

$\ldots = kx^{\frac{1}{2}} \ln x \pm \int f(x),$ *with f(x) not involving the 'original' ln x*

A1

$\ldots = 2x^{\frac{1}{2}} \ln x - 4x^{\frac{1}{2}} \, (+c)$

Condone absence of '+ c'

A1

3

(c) $\int_0^{e} \frac{\ln x}{\sqrt{x}} dx = \lim_{a \to 0} \int_0^{e} \frac{\ln x}{\sqrt{x}} dx$

M1

$$= -2e^{\frac{1}{2}} - \lim_{a \to 0}\left[2a^{\frac{1}{2}}\ln a - 4a^{\frac{1}{2}}\right]$$

F(b) – F(a)

M1

But $\lim_{a \to 0} a^{\frac{1}{2}}\ln a = 0$

Accept a general form e.g $\lim_{x \to 0} x^k \ln x = 0$

B1

So $\int_0^e \frac{\ln x}{\sqrt{x}}$ dx exists and = $-2e^{\frac{1}{2}}$

A1

4

[8]

Q23.

(a) $\int\left(x^{\frac{1}{3}} + x^{-\frac{1}{3}}\right)dx = \frac{3}{4}x^{\frac{4}{3}} + \frac{3}{2}x^{\frac{2}{3}}\ (+c)$

M1 for adding 1 to index at least once

M1A1

$\int_0^1 \ldots = \left(\frac{3}{4} + \frac{3}{2}\right) - 0 = \frac{9}{4}$

Condone no mention of limiting process;
m1 if "– 0" stated or implied

m1A1

4

(b) Second term is $x^{-\frac{4}{3}}$

B1

Integral of this is $-3x^{-\frac{1}{3}}$

M1 for correct index

M1A1

$x^{-\frac{1}{3}} \to \infty$ as $x \to 0$, so no value

E1

4

[8]

Q24.

(a) 0

B1

1

(b) $u = xe^{-x} + 1 \Rightarrow du = (e^{-x} - xe^{-x})\,dx$

Attempts to find du

M1

$$\int \frac{e^{-x}(1-x)}{xe^{-x}+1}\,dx = \int \frac{1}{u}\,du = \ln u + c = \ln(xe^{-x}+1)\{+c\}$$

Condone missing c

A1

2

(c) $$\int \frac{1-x}{x+e^{x}}\,dx = \int \frac{e^{-x}(1-x)}{xe^{-x}+1}\,dx$$

B1

$$\int_1^{\infty} \frac{1-x}{x+e^{x}}\,dx = \lim_{a\to\infty}\left[\ln(xe^{-x}+1)\right]_1^a$$

$$= \lim_{a\to\infty}\left\{\ln\left(ae^{-a}+1\right)\right\} - \ln\left(e^{-1}+1\right)$$

For using part (b) and $F(B) - F(A)$

M1

$$= \ln\left\{\lim_{a\to\infty}\left(ae^{-a}+1\right)\right\} - \ln\left(e^{-1}+1\right)$$

$$= \ln 1 - \ln\left(e^{-1}+1\right) = -\ln\left(e^{-1}+1\right)$$

For using limiting process

M1
A1

4

[7]

Q25.

(a) (i) $\int x^{-\frac{1}{2}}\,dx = 2x^{\frac{1}{2}}(+c)$

M1 for $kx^{\frac{1}{2}}$

M1A1

$\int_0^9 \frac{1}{\sqrt{x}}\,dx = 6$

ft wrong coeff of $x^{\frac{1}{2}}$

A1ft

3

(ii) $\int x^{-\frac{3}{2}}\,dx = -2x^{-\frac{1}{2}}(+c)$

M1 for $kx^{-\frac{1}{2}}$

M1A1

$x^{-\frac{1}{2}} \to \infty$ as $x \to 0$, so no value

'Tending to infinity' clearly implied

E1

3

(b) Denominator $\to 0$ as $x \to 0$

E1

1

[7]

Q26.

(a) $\int xe^{-2x}\,dx = -\frac{1}{2}xe^{-2x} - \int -\frac{1}{2}e^{-2x}\,dx$

M1

Reasonable attempt at parts

A1

$= -\frac{1}{2}xe^{-2x} - \frac{1}{4}e^{-2x}\,\{+c\}$

Condone absence of + c

A1ft

$\int_0^a xe^{-2x}\,dx = -\frac{1}{2}ae^{-2a} - \frac{1}{4}e^{-2a} - (0 - \frac{1}{4})$

F(a) − F(0)

M1

$\frac{1}{4} - \frac{1}{2}ae^{-2a} - \frac{1}{4}e^{-2a}$

A1

5

(b) $\lim_{a \to \infty} a^k e^{-2a} = 0$

B1

1

(c) $\int_0^\infty xe^{-2x}\,dx$ =

$\lim_{a \to \infty}\{\frac{1}{4} - \frac{1}{2}ae^{-2a} - \frac{1}{4}e^{-2a}\}$

If this line oe is missing then 0/2

M1

$= \frac{1}{4} - 0 - 0 = \frac{1}{4}$

On candidate's "1/4" in part (a).
B1 must have been earned

A1ft

2

[8]

Q27.

(a) $\Rightarrow \lim_{a\to\infty}\left(\frac{3+\frac{2}{a}}{2+\frac{3}{a}}\right) = \frac{3+0}{2+0} = \frac{3}{2}$

M1 A1

2

(b) $\int_1^\infty \frac{3}{(3x+2)} - \frac{2}{2x+3}\,dx$

$= [\ln(3x+2) - \ln(2x+3)]_1^\infty$

M1

*a*ln*(3x + 2) +* *b*ln*(2x + 3)*

A1

$$= \left[\ln\left(\frac{3x+2}{2x+3}\right)\right]_1^\infty$$

m1

$$= \ln\left\{\lim_{a\to\infty}\left(\frac{3a+2}{2a+3}\right)\right\} - \ln 1$$

M1

$$= \ln\frac{3}{2} - \ln 1 = \ln\frac{3}{2}$$

CSO

A1 5

[7]

Q28.

(a) $\int x^{-3}\, dx = kx^{-2}\,(+c)$

M1

$x^{-2} \to 0$ as $x \to \infty$

M1

Improper integral has value 1

A1 3

(b) No value as *x* term tends to ∞

OE

B1 1

(c) $\int x^{-2}\, dx = kx^{-1}\,(+c)$

M1

$x^{-1} \to 0$ as $x \to \infty$

m1

Improper integral has value 5

A1 3

[7]

Examiner reports

Q3.

Those students who wrote the integrand in the form $x^{-1.5}$ normally presented a correct solution although some did not adequately deal with ∞ as a limiting case. There were a number of incorrect solutions seen with terms which included ln x, and these scored no marks.

Q4.

Explanations for why the integral was improper were slightly better than those given for the corresponding question in 2012, but there are still a number of students whose explanations are not sufficiently precise. Use of the word ‘it’ instead of, for example, ‘integrand’, when explaining what is not defined at the lower limit, 0 illustrates the imprecision. In general the method of integration by parts was applied accurately and a higher proportion of students showed the relevant limiting process than last year.

Q5.

Most students applied the correct method by multiplying out the bracket before integrating but there were a surprisingly many who had sign errors in their later work. In contrast there were some excellent solutions seen which involved the ‘Lim’ notation correctly applied.

Q6.

This question asked students to evaluate an improper integral, showing the limiting process used. The majority of students correctly equated the given integral to $\lim_{a\to\infty}\int_0^a\left(\frac{2x}{x^2+4}-\frac{4}{2x+3}\right)$ dx and most students integrated correctly to obtain the difference of two logarithms although an inverse trigonometric function was seen occasionally. A significant minority of students then dealt with the upper limit incorrectly by considering each logarithmic term separately and stating that each was ∞ so the limit of their difference was 0. More able students combined the two terms into a single log term, substituted the limits and reached $\lim_{a\to\infty}\left[\ln\left(\frac{a^2+4}{(2a+3)^2}\right)\right]-\ln\left(\frac{4}{9}\right)$. Further work, which was required on the first term before the limit taken, was not always shown. Those students who squared the denominator and then divided each term in the relevant fraction by a^2 to obtain $\lim_{a\to\infty}\left[\ln\left(\frac{1+\frac{4}{a^2}}{4+\frac{12}{a}+\frac{9}{a^2}}\right)\right]-\ln\left(\frac{4}{9}\right)$ usually obtained the correct answer convincingly.

Q7.

The basic integration was usually carried out correctly in this question on improper integrals, but a significant minority of candidates failed to include reference to the limit of (3)$x^{1/3}$ as $x\to\infty$ within their concluding statement in part (a).

In part (b) a surprising number of solutions ended with the value $-\frac{3}{2}$ instead of the correct value $\frac{3}{2}$.

Q8.

Explanations for why the integral was improper were frequently less than convincing. The examiners were ideally looking for the response 'The interval of integration is infinite. '

In part (b) a significant minority could not reach the correct integral in terms of u. Errors of the type $\frac{1}{x^2+3e^{4x}}$ = $x^{-2} + 3e^{-4x}$, seen in a number of solutions, were disappointing.

In part (c), the improvement in answers to questions on this topic seen in recent question papers was not maintained. There was a smaller proportion of candidates starting off correctly by, for example, writing $\int_{\frac{1}{2}}^{\infty} \frac{x(1-2x)}{x^2+3e^{4x}}\,dx = \lim_{a \to \infty} \int_{\frac{1}{2}}^{a} \frac{x(1-2x)}{x^2+3e^{4x}}\,dx$.

Q9.

Part (a) was, as expected, generally answered well, with students clearly showing the method of integration by parts. In part (b), a higher proportion of students than in previous papers replaced the upper limit with a constant (for example, a) indicated that $a \to \infty$, and then found $F(a) - F(0)$. However, a significant number of students were not explicit in indicating that the limit as $a \to \infty$, $a^k e^{-a} = 0$ and hence they lost the final two marks. This is illustrated by the common error which was just to state that as $a \to \infty$, $e^{-a} \to 0$ so given integral = 2.

Q10.

This question again provided a good number of marks for the vast majority of candidates, who showed an adequate knowledge of integration, but in an unexpectedly large number of cases sign errors were made in part (a), causing the letters p and q to be interchanged. In part (b), most candidates were able to identify correctly which integral had the finite value.

Q11.

Most candidates found the correct value for C in part (a). A large majority of candidates realised that part (a) had some relevance to part (b) and duly wrote the integrand in terms of partial fractions and generally integrated correctly. Those who missed this step gained little or no credit for their later work. Although showing the limiting process is an area of the specification that candidates still find difficult, overall improvement continues to be noted.

Q12.

Part (a) was, as expected, very well answered with candidates clearly showing the method of integration by parts. In part (b), a significant number of candidates were not explicit in explaining that the integrand was not defined at $x = 0$. It was pleasing to see in part (c) that more candidates than usual replaced the lower limit by, for example, a, carried out the integration and then considered the limiting process as $a \to 0$. However, a significant number of candidates did not specifically consider $\lim_{a \to 0} a^3 \ln a$.

Q13.

Most candidates showed confidence with the integration needed in this question but were much less confident with the concept of an improper integral. The explanations in part (a) were often very wide of the mark, and indeed quite absurd, while in other cases the statements made were too vague to be worthy of the mark, using the word "it" without making it clear whether this referred to the integrand or to the integrated function. Another mark was often lost at the end of the question, where candidates thought that $0^{-\frac{1}{4}}$ was equal to zero.

Q14.

In part (a), a significant number of candidates failed to give the correct reason for why the integral was improper, with most giving the reason that the integrand was undefined for the upper limit rather than that the interval of integration was infinite. In part (b)(i), a large number of candidates failed to use the substitution correctly. Part (b)(ii) was generally answered well with most candidates integrating by parts correctly to score 3 of the 5 marks available, but some candidates did not gain the final two marks as examiners did not see the lower limit replaced by, for example, a and the consideration of the limiting process as $a \to 0$. Part (b)(iii) was answered incorrectly by the majority of candidates, who usually either stated the same value or the reciprocal value of their answer to part (b)(ii).

Q15.

Improper integrals and limiting processes continue to cause problems for a significant minority of candidates. Part (a) was generally not answered well with too many candidates making a statement which they then contradicted in part (c). As highlighted in the MFP3 examiners' reports for January 2008 and January 2010, an integral, with ooas a limit, is improper because the interval of integration is infinite.

Most candidates correctly applied integration by parts to score the three marks available in part (b).

In part (c) examiners expected to see the infinite upper limit replaced by, for example, a, the integration carried out and then consideration of the limiting process as $a \to \infty$.

It was not uncommon to see in solutions the statement 'as $a \to \infty$, $e^{-4a} \to 0$ so $e^{-4a}(-a - 1/4) \to 0$' without seemingly any particular analysis of $\lim_{a \to \infty} a\, e^{-4a}$. This lack of analysis was penalised.

Q16.

Most candidates were able to integrate the given powers of x, and many were able to draw the correct conclusions, though sign errors were common in part (b). In part (c), most of those who had been successful so far were able to complete the task successfully, but it was noticeable that some candidates did not take a hint from the award of only one mark for this part of the question: instead of drawing a quick conclusion from the results already obtained they went through the whole process of integrating, substituting and taking limits, possibly losing valuable time.

Q17.

Part (a) was, as expected, very well answered with candidates clearly showing the method of integration by parts. In part (b), a smaller proportion of candidates than in June 2008 showed the limiting process fully. For full marks candidates were also expected to pay particular attention to the value of the limit of (a ln a) as a tended to 0.

Q18.

This question on improper integrals and limiting processes, which lacked the structure given in many previous papers, was the worse answered question on the paper. Examiners expected to see the infinite upper limit replaced by, for example, a, the integration then carried out and then consideration of the limiting process as $a \to \infty$. Most, although not all, reached 'ln x – ln($4x$ + 1)' for 1 mark, but many of the weaker candidates just stated that ln x – ln($4x$ + 1) = 0 when $x \to \infty$ and gave the wrong value 'ln 5' as their answer. Better candidates scored 3 marks for reaching as $a \to \infty$, $\ln\left(\frac{5a}{4a+1}\right) = \ln\frac{5}{4}$', but only those who inserted the extra step to get, for example, 'as $a \to \infty$, $\ln\left(\frac{5}{4+\frac{1}{a}}\right) = \ln\frac{5}{4}$', were in line for full marks.

Q19.

Part (a) was generally not well answered with a significant minority either not attempting it or making a statement which they then contradicted in part (c). The method of integration by parts was understood with the great majority obtaining the correct answer to part (b). Although there continues to be an improvement in candidates' solutions to the evaluation of an improper integral, there were still a significant minority who made no attempt to show the limiting process used.

Q20.

There were many all-correct solutions to this question from the stronger candidates. Others were unsure of themselves when dealing with the behaviour of powers of x as x tended to infinity. Many others did not reach the stage of making that decision: either they failed to convert the integrands correctly into powers of x, or they integrated their powers of x incorrectly. A reasonable grasp of AS Pure Core Mathematics is essential for candidates taking this paper.

Q21.

Most candidates applied integration by parts accurately to obtain the correct answer for the integral of x^3 ln x. Although many candidates gave a correct explanation in part (b) for why the integral was improper, there were others whose incorrect explanations centred on the limit e or the interval of integration being infinite. For full marks in part (c), candidates were expected to pay particular attention to the value of the limit of, for example, a^4 ln a as a tended to 0.

Q22.

A significant minority did not attempt part (a), or used the limit e rather than 0 in their explanations of why the definite integral was improper. Using integration by parts in part (b) proved to be more of a problem for candidates than expected. Although a higher proportion of candidates showed the limiting process than in previous sessions, this topic still remains a section of the specification which is generally not fully understood.

Q23.

This question proved to be very largely a test of integration. Many candidates answered part (a) without any apparent awareness of a limiting process, but full marks were awarded if the answer was correct. The positive indices meant that the powers of x would tend to zero as x itself tended to zero.

In part (b) a slightly more difficult integration led to one term having a negative index. Three marks were awarded for the integration, but for the final mark it was necessary to give some indication as to which term tended to infinity. Many candidates did not gain this last mark, but it was sad to see how many did not gain any marks at all in part (b). This was usually for one of two reasons. Either they failed to simplify the integrand and carried out a totally invalid process of integration, or they saw that the denominator x of the integrand would become zero and said that this made the integral 'improper' and therefore incapable of having any value. Since it was stated in the question that **both** integrals were improper, this comment failed to attract any sympathy from the examiners.

Q24.

The standard result required in part (a) was known by most candidates. In part (b), a minority of candidates failed to use the given substitution and a significant minority of others left their final answer in terms of u. In part (c) it was not always clear to the examiners that a candidate had fully seen the relationship between the integrands in parts (b) and (c). Some candidates clearly showed the relationship by multiplying both the numerator and the denominator of the integrand in part (c) by e^{-x}. Sometimes poor use of brackets led to the wrong answer '2' presumably from $-\ln e^{-1} + 1$. However it was pleasing to see the increased number of candidates who correctly used the limiting process this year.

Q25.

There were many faults in the attempts at integration in this question, especially in part (a)(ii). Those who succeeded in the integration often failed to obtain full credit as they did not realise that $0^{-\frac{1}{2}}$ had no finite value. Part (b) was not well answered, even thosewho had succeeded in part (a)(ii) often failing to identify the feature that made the integrals 'improper'.

Q26.

Most candidates applied integration by parts correctly in part (a) but it was disappointing to see a significant minority not attempting to substitute the given limits. The standard result required in part (b) was well known. In the final part, although many showed the limiting process used, a significant number of candidates just replaced x by ∞ without comment.

Q27.

Examiners expected to see the limiting process used in part (b). In part (a) common errors included substituting a ‘large’ number, usually a million, for a or writing $\frac{3\infty}{2\infty}$.

It is worth noting that the solution $= \lim_{a \to \infty} \frac{3a}{2a+3} + \frac{2}{2a+3} = \frac{3}{2}$ since ‘$= \lim_{a \to \infty} \frac{2}{2a+3} = 0$’ is insufficient without due consideration of ‘$= \lim_{a \to \infty} \frac{3a}{2a+3}$’.

In part (b) it was surprising to see candidates giving the answer ln(3x + 2) –21n (2x + 3)’ for the indefinite integral. A significant minority did not include any limiting process in their evaluation of the definite integral.

Q28.

Many answers to this question were spoiled by poor integration. However, many candidates were still as able to show some understanding of the significance of the negative power of x tending to zero as x tended to infinity. Answers to part (b) tended to be vague, but credit was given if the candidate had an x term not present in part (a) and then stated that the integral had no finite value. In part (c), many candidates revealed their lack of experience of integration by simply integrating both factors and multiplying the results.